

Comparing the efficacy of Rapid Review with a Systematic Review in the Software Engineering Field

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ABSTRACT

Context: Rapid Reviews are secondary studies aiming to deliver evidence to experts in a more timely manner and with lower costs than traditional literature reviews. Previous studies have shown that experts and researchers are positive toward Rapid Reviews. However, little is known about how Rapid Reviews differ from traditional Systematic Reviews.

Objective: The goal of this paper is to compare a Rapid Review with a Systematic Review in terms of their methods (e.g., search strategy, study selection, quality assessment, data extraction) and findings to understand how optimizing the traditional Systematic Review method impacts what we obtain with Rapid Review.

Method: To achieve this goal, we conducted a Systematic Review with the same research questions answered by a pre-existing Rapid Review and compared those two studies. Also, we surveyed experts from industry and academia to evaluate the relevance of the findings obtained from both the secondary studies.

Results: The Rapid Review lasted six days, while the Systematic Review took one year and two months. The main bottlenecks we identified in the Systematic Review are (i) executing the search strategy and (ii) the selection procedure. Together they took ten months. The researchers had to analyze the information from 11,383 papers for the Systematic Review compared to 1,973 for the Rapid Review. Still, most (~78%) of the papers included in the Systematic Review were returned by the Rapid Review search, and some papers that could be included were unduly excluded during the Rapid Review's selection procedure. Both secondary studies identified the same number of pieces of evidence (30), but the pieces of evidence are not the same.

Conclusion: The Rapid Review and Systematic Review results are inherently different and complementary. The time and cost to conduct a Systematic Review can be prohibitive in experts' contexts. Thus, at least in such situations, a Rapid Review may be an adequate choice. Moreover, a Rapid Review may be executed in the experts' context as a previous low-cost step before deciding to invest in a high-cost Systematic Review.

1. Introduction

Evidence-Based Software Engineering was introduced in the research community to integrate the best research evidence with practice [1]. Moreover, it is argued that the knowledge aggregated with secondary studies is the most appropriate to be transferred to practice [2].

There are several methods for conducting a secondary study; however, the method choice must be determined by the research objective. The most commonly used are Systematic Mapping Studies and Systematic Reviews (SRs). Despite their similarities, there are differences in terms

of purpose and approach [3]. Typically, Systematic Mapping Studies have the goal of classifying and thematically analyzing the literature on a topic with a broad scope, while Systematic Reviews have the goal of identifying best practices related to methods, tools or technologies, with a focused scope and related to empirical evidence [4]. Therefore, they tend to focus on practical problems. However, some researchers observed that secondary studies — notably Systematic Reviews — lack connection with Software Engineering practice due to many reasons, for instance, the amount of time and effort to conduct them [5, 6], thereby not producing timely results for experts.

Rapid Reviews (RRs) [7] are one of the alternatives to reduce the gap between research and practice. They are secondary studies that provide scientific evidence to support experts in making evidence-informed decisions, and usually are conducted in a wider knowledge transfer initiative [8, 9]. Rapid Reviews adapt Systematic Reviews' method to convey evidence through more appealing mediums [10, 11], reduced time frames, and lower costs. To make that possible, Rapid

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Reviews simplify Systematic Reviews method by adhering to strategies such as: limiting search sources, having just one person to screen studies, skipping quality appraisal, among others [12].

Although more than 100 Rapid Reviews are reported in the medical field [13, 14], the Software Engineering community has not adopted Rapid Reviews as extensively. In 2018, Cartaxo et al. [7] reported the first Rapid Review in Software Engineering. Since then, software engineering researchers have used Rapid Reviews on different topics, such as Amalfitano et al. [15] explore aspects related to testing of context-aware software systems; De Souza et al.'s [16] investigate how to capture and represent smartness regarding IoT software systems; Ponce et al. [17] overlook strategies to migrate from a monolithic architecture to microservices. Meanwhile, there are also initiatives to evaluate how Rapid Reviews could be applied in different scenarios. For instance, Pizard et al.'s work [18] about using Rapid Reviews for Evidence-Based Software Engineering training, as well as another paper from the same group, about the adoption of Rapid Reviews in governmental agencies [19]. In addition, Rico et al. [20] proposed a guideline for conducting Interactive Rapid Reviews (IRR) in software engineering, shifting from a focus on technology transfer to knowledge exchange.

In this context, we believe it is important not only to have more Rapid Reviews in Software Engineering but also to investigate their intricacies. For instance, a recent study [21] explored the perceptions of the Software Engineering research community regarding Rapid Reviews. One of the findings is that a large group of researchers might adopt Rapid Reviews as a knowledge transfer method. However, researchers with such a point of view emphasize the importance of studies contrasting Rapid Reviews with Systematic Reviews. They argue that it is necessary to understand the impact of simplify the traditional Systematic Reviews' method and enable an informed decision on whether or not to conduct a Rapid Review and in which context.

Therefore, the **GOAL** of this research is to compare a Rapid Review with a Systematic Review in terms of their methods and findings in order to understand how optimizing the traditional Systematic Review method impacts what we obtain with Rapid Reviews.

To achieve this goal, we conducted a Systematic Review with the same Research Questions of a Rapid Review we have previously published [7] and performed the comparison between those two studies. In this paper, we report that comparison, not the Systematic Review itself. In addition, we conducted a survey with experts from industry and academia to evaluate the relevance of the findings we obtained with both the secondary studies.

We acknowledge that the purpose of both methods (Systematic Review and Rapid Review) are different and complementary [7]. Thus, the intention is not to allege that one is better than the other. Also, all claims we make throughout this paper should be taken with due care. As

mentioned earlier, there is a need to compare Rapid Review and Systematic Review methods, and this study is the first step toward understanding the Rapid Review or Systematic Review methods selection.

Although there are some studies comparing the two approaches in medicine [22, 23, 24], this is the first study in the Software Engineering field. We also dived more in-depth with our comparison – enabling us to present more detailed insights – when considering the comparisons between Rapid Reviews and Systematic Reviews in health-related research fields. Section 2 details how our comparison is more thorough than the ones in medicine.

The main **contributions** of this paper are:

- We demonstrate where in the review process a Systematic Review consumes more effort and duration relative to a Rapid Review;
- We demonstrate that a Rapid Review provides evidence that is sufficiently relevant for practice, compared to a Systematic Review;
- We demonstrate that a Rapid Review does make equivalent findings compared to the Systematic Review;
- We mapped each secondary study's decisions and data to explain the reason for each difference between those two studies' findings;
- We guide when to conduct a Rapid Review and when to conduct a Systematic Review.

2. Previous Work Comparing Systematic Reviews with Rapid Reviews

To the best of our knowledge, this is the first paper reporting a comparison between a Systematic Review and a Rapid Review in the Software Engineering context. In this section, we present comparative studies conducted in health research fields.

Taylor-Phillips et al. [23] conducted a study to compare a Systematic Review to two types of Rapid Reviews, one with all steps optimized and another with fewer steps optimized. The comparison focused on the difference between the studies included in each secondary study and the difference between their findings. The effort to achieve the results also was considered by the number of studies returned by the search. They could not find differences between the Systematic Review and the Rapid Review with fewer steps optimized. However, the Rapid Review with all steps optimized had increased biases and could not identify papers that influenced the findings.

Watt et al. [25] compared the conclusions produced by Rapid Reviews and Systematic Reviews. The purpose was to identify how Rapid Reviews are conducted in medicine compared to Systematic Reviews on the same topic. The

time frame and the conclusions were taken into consideration. The study claims that the results were not essentially different.

On the other side, Van de Velde et al. [24] compared the findings of their Rapid Review with a Systematic Review conducted by a different research group that published on the same topic, the medicinal use of potato-derived products, and conflicting results were observed between the secondary studies.

The Systematic Review identified two trials about potato peels in treating burns but concluded that honey dressings are better than potato peel dressings. The Rapid Review found additional evidence indicating that potato peel dressings are better than gauze alone. The comparative study concludes that although potato peel dressings as not applicable to Western countries' settings can be a good alternative for emerging countries.

The comparisons between Rapid Reviews and Systematic Reviews in medicine focus mainly on the differences between the findings they derived and the studies they included. In this research, we take a step further and dive into detail, comparing each aspect of a Rapid Review with a Systematic Review. More importantly, we mapped each secondary study's decisions and data to explain the reason for each difference between those two studies. For instance, we could track that a specific finding was identified in one secondary study but not in the other due to a particular term in one of the search strings. Similarly, we could determine that a specific finding identified in one secondary study was not found in the other because the paper that provided that finding was excluded on the first step of the selection procedure.

In addition to the fact that our comparison approach is more in-depth than those in medicine, it is also relevant to highlight that Rapid Reviews and Systematic Reviews in Software Engineering may differ considerably from those in health research fields.

In the Software Engineering context, secondary studies tend to be more qualitative-driven. Most Software Engineering research needs to consider human aspects inherent to the software development process. Therefore, the synthesis procedure, and consequently the results, tend to suffer from biases related to researchers' subjectivity. This usually does not happen in medicine, where secondary studies tend to synthesize evidence from randomized controlled trials, following meta-analysis principles.

Regarding Rapid Reviews, one difference between Software Engineering and medicine is their time frames and criticality level. Rapid Reviews in Software Engineering are usually applied in an ever-changing context and, in general, not in a critical setting. This is the opposite of what occurs in medicine. Cartaxo et al. [12] mentioned that a Rapid Review in Software Engineering can be conducted in a few weeks, while in medicine, it usually takes a few months.

Those differences directly influence the way Rapid Reviews are conducted in Software Engineering and medicine. That is why we believe a comparison like the one we present

in this paper is important, even considering that there are already other comparisons in health research fields.

3. Method

We conduct this research in three main steps. First, we perform a Systematic Review with the same scope as one of our pre-existing Rapid Reviews. Then, in the second step, we compare those two secondary studies regarding the method they followed and the findings they obtained, both from qualitative and quantitative perspectives. And finally, in the third and last step, we conduct a survey with experts from industry and academia, aiming to understand their perceptions concerning the relevance of the findings we obtained with both the Rapid Review and the Systematic Review.

In order to achieve our objective, we will strive to provide answers to the following Investigative Research Questions (IRQs):

- **IRQ1:** What factors explain the difference in effort between a Systematic Review and a Rapid Review?
- **IRQ2:** What factors explain the difference in relevance between the findings of a Systematic Review and a Rapid Review?
- **IRQ3:** Are the findings from a Systematic Review and a Rapid Review significantly different in some ways?

3.1. The Rapid Review

At the time we conducted this research, there were few Rapid Reviews published in Software Engineering peer-review venues. So, we decided to select a Rapid Review conducted by our research group [7]. By following this strategy, we also have access to Rapid Review details that enable us to make a richer comparison.

The Rapid Review was conducted in the context of a software project in a software development company. The project team adhered to agile practices but was facing problems regarding low customer collaboration. And, addressed that situation, providing evidence that helped the experts to better understand their problem and, especially, providing strategies to improve customer collaboration.

3.2. The Systematic Review

Since our goal is to compare a Systematic Review with a Rapid Review, not only in terms of their findings but also their methods, we started from the same Research Questions answered by the Rapid Review, to make both studies comparable. Nevertheless, from that point on, the Systematic Review follows a separated path. To guarantee such independence, the leading researcher of the Systematic Review did not read or had any contact with the Rapid Review in any way.

We decided that the Systematic Review and the Rapid Review should follow a different path as early as before creating their protocols. The rationale behind this decision is twofold. First, although we adopted the same research questions on both the Rapid Review and the Systematic

Table 1

Aspects analyzed to compare the Systematic Review with the Rapid Review.

Aspects	Items
Motivating Problem	The problem that motivated conducting the secondary studies.
Protocol	The procedures documented in the protocol and the level of detail.
Time Frame	The time frame demanded to conduct the secondary studies, and of each of their procedures.
Search Strategy	The search sources; the structure and terms of the search strings; the time span of the searched papers; and the number of papers found with the searches.
Study Selection	The steps composing the selection procedure; how many and who is involved in each step; the number of studies excluded in each step; how many and which studies are included in the end; and, in which step a study included in one secondary study was improperly excluded from the other.
Quality Appraisal	How the quality appraisal was conducted; who was involved in this procedure; and the number of papers excluded due to quality criteria.
Data extraction	How data extraction procedure was conducted and who was involved in it.
Synthesis	How the synthesis procedure was conducted, and who was involved in it.
Findings	The findings and the reasons for some findings to show up in one secondary study but not in the other.
Report	The medium which the findings are presented and the level of detail.

Review, the Systematic Review protocol was created by a researcher who was not part of the Rapid Review team to avoid bias due to the learning effect. Second, creating the Systematic Review protocol from scratch also reveals what can happen differently when one has less time pressure and no close contact with the experts creating the protocol. The opposite of what happened with the Rapid Review's leading researcher when creating the protocol.

Since the goal of this study is not the Systematic Review itself but its comparison with the Rapid Review, we do not present the entire protocol of the Systematic Review here. Although, in Section 4 we discuss the Systematic Review steps in detail.

3.3. Comparing the Rapid Review with the Systematic Review

To conduct the comparison between the two secondary studies, we analyzed each aspect described in the Systematic Reviews' [26] and Rapid Reviews' [12] guidelines. We also take into account the comparisons between Systematic Reviews and Rapid Reviews in the health field. In summary, we compared the aspects listed in Table 1.

3.4. Survey with Experts

After obtaining the Systematic Review results, we administered a survey aiming to investigate how relevant the findings of the Systematic Reviews and Rapid Reviews are from the perspective of experts from industry and academia. A draft of the survey was created by three researchers and administered as a pilot to the other three researchers, which provided us with comments to improve it.

The recruitment was based on convenience sampling [27], inviting potential participants via email. The participants were informed that the survey was anonymous, for research purposes only, and they could withdraw their participation anytime they wish [28]. Then, we sent the online survey in Google Forms format to the participants.

The participants were also asked to invite any colleague who would fit the required profile. The profile we searched for is either a person working in the industry who has software development experience. Altogether, 39 participants answered our survey.

The survey is composed of two sections. In the first section, there are demographic questions to identify participants' age, level of educational attainment, years of professional experience, current job role, and if they currently work in academia, industry, or both. We decided to keep the demographic section as short as possible, since the second section is somewhat extensive, although very straightforward.

In the second (main) section, we provide a brief context about the problem and the four research questions that drive both our Rapid Review and Systematic Review, which are related to customer collaboration in software development teams. Then, we ask the participants to classify, on a five-point Likert scale, how relevant is each piece of evidence we have found in the two secondary studies. For example, one of Rapid Review's pieces of evidence that answers RQ1, about the benefits of good customer collaboration, is: *It leads to greater productivity*. Another example is Systematic Review's piece of evidence that answers RQ2, concerning problems caused by low customer collaboration: *It causes misunderstandings*.

4. Results Comparing the Rapid Review with the Systematic Review

In this section, we present the convergences and divergences comparing the two secondary studies regarding the method they followed and the findings they obtained, both under qualitative and quantitative perspectives.

4.1. Motivating Problem

The problem that motivated the conduction of the Rapid Review emerged from a software development project in a software company. The experts reported having problems related to low customer collaboration. They would like to have access to evidence that could mitigate that problem, which in turn motivated the conduction of the Rapid Review. Prior to conducting the Rapid Review, the researchers contacted representatives of the development team to discuss problems related to low customer collaboration and to obtain the entire context related to the problem faced by the organization.

We followed the same motivation for conducting the Systematic Review. We conducted the review aiming to find evidences about issues about customer collaboration in software development. We also wanted to know if the set of findings would be similar or different to the Rapid Review findings.

4.2. Research Questions

The Rapid Review's research questions were elaborated in close collaboration with experts aiming to provide evidence to help them make informed-decisions in practice. The research questions are:

- **RQ1:** What are the benefits of customer collaboration in software development practice?
- **RQ2:** What are the problems caused by low customer collaboration in the software development practice?
- **RQ3:** What are the challenges in establishing customer collaboration in software development practice?
- **RQ4:** What are the strategies to improve customer collaboration in software development practice?

The Systematic Review's research questions are the same as those in Rapid Review, aiming to enable a full comparison between them. So, the research questions were our controllable variable.

4.3. Protocol

The Rapid Review's protocol¹ has the following main sections: practical problem, research questions, search strategy, selection procedure, quality assessment, extraction procedure, synthesis procedure, and review report. Although there is a specific section for quality appraisal, it only declares that no quality appraisal was conducted. The document contains succinct descriptions of each step but lacks detailed information on how the procedures were performed as well as theoretical justification supporting its technical

¹RR's protocol: <https://bit.ly/3FzwwiZ>

decisions. Because of this, it has three pages. A single researcher created the protocol and the experts reviewed it to guarantee the Rapid Review would be aligned with their needs.

The Systematic Review's protocol² includes the following main sections: introduction, background, research questions, search strategy, selection process, quality analysis, data extraction, data synthesis, and limitations. The document details each step of the Systematic Review method as recommended by Kitchenham et al. [1, 26]. The research decisions are supported by references to scientific studies.

4.4. Stakeholders Role

Although only one researcher led the entire Rapid Review process, it was conducted in close collaboration with experts who were immersed in a practical problem. This is not usually observed in Systematic Reviews [6]. In addition, the protocol of the Rapid Review does not describe how many experts collaborated with the conduction. Thus, we need take a look at the paper that reports the Rapid Review [7], which says that five experts were involved.

In the Systematic Review protocol, on the other side, we could find only bibliographic discussions aiming to justify the review's execution. In addition, to avoid bias, nine researchers participated in the paper selection procedure, and some other steps (e.g., quality analysis, extraction procedure, and synthesis procedure) were revised by at least one researcher.

Both researchers who led each of the secondary studies are from the same software engineering research group.

4.5. Time Frame

Regarding the duration of the two secondary studies, the results are quite contrasting. Figure 1 summarizes the time frame of both the Systematic Review and the Rapid Review.

The Rapid Review's planning, execution, and reporting took approximately six days of dedication of a single full-time researcher. The process was so short, and the activities so tangled in one another that it was not possible to measure the duration of each activity independently.

The Systematic Review's planning, execution, and reporting spanned a year and two months. The Systematic Review's leading researcher has worked continuously on it, but the dedication time was not tracked fine-grained enough to measure the effort in hours or days, as with the Rapid Review.

Although we could not investigate the time frames of each Rapid Review's steps in detail, we could do that for the Systematic Review. This is important because it shows the main bottlenecks of the Systematic Review method, revealing why such a study takes so much time and effort when compared to a Rapid Review. Following we detail the duration of each of the Systematic Review's steps:

- **Creating the Protocol [2 months]:** We created the protocol from scratch, not based on the Rapid Review's one.

²SR's protocol: <https://bit.ly/4dBx7DM>

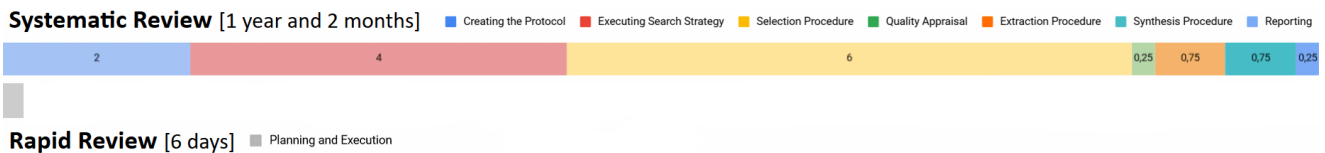


Figure 1: Time frames of both the secondary studies. The numbers in the systematic review time frame are the duration of each activity in months. The rapid review has no detailed time frame for each activity due to its low duration and tangled nature.

The first author defined and edited the protocol, which has been revised several times by three researchers, two senior researchers and a PhD student. The various back-and-forth inherent to the revision process and the researchers' busy agendas are the main reasons for the time we spent creating the protocol.

- **Executing the Search Strategy [4 months]:** Three months and three weeks for manual search, and one week for automatic search. This period includes extracting the metadata of all found papers and organizing it on the selection spreadsheets. Such activity is very tedious and time-consuming. The manual search required more effort because it involved visiting the manual databases defined in Table 3 and checking each of their issues, from the creation of the database until the date of the Rapid Review. Metadata such as the title, summary, authors, and link of each paper were extracted and recorded in a spreadsheet. After completing the manual search and obtaining a full spreadsheet with individual metadata from 8,696 papers, we combined these manual results with the 3,601 papers returned by the automatic search. The automatic search, on the other hand, could be partially supported by some search engines that enabled us to extract search results metadata in a very straightforward way. The first author executed this steps under the supervision of a senior researcher.
- **Study Selection [6 months]:** The Systematic Review's first selection step (i.e., title and abstract) lasted three months and a half. Although analyzing just papers' titles and abstracts is straightforward, the high number of papers (11,383 non-duplicated) returned by the search sources prolonged this step. The second selection step (i.e., complete paper content) lasted two and a half months. One of the major causes of delay occurred due to the dependency on researchers outside the research group to conduct the selection procedure in pairs. It involved seven researchers with limited availability. Finding a common schedule for the conflict meetings with the pairs was difficult.
- **Quality Assessment [0.25 month]:** This step was quite straightforward, taking just one week to analyze 63 papers. This step was reviewed by a senior researcher.
- **Data Extraction [0.75 month]:** It took three weeks, and it includes creating and filling a spreadsheet with the data extracted to answer each of the research questions from the 63 included papers. This step was reviewed by a senior researcher.

- **Synthesis [0.75 month]:** It took three weeks to apply the thematic analysis and synthesize the extracted data. This step was reviewed by a senior researcher.
- **Reporting [0.25 month]:** It took one week to write down the document reporting the results of the Systematic Review. This step was reviewed by a senior researcher.

4.6. Search Strategy

The Rapid Review adopts a single strategy (automated search only) and a single search source (one search engine only) approach. The goal is to make the search process more straightforward. The single search source is the Scopus search engine.

The Systematic Review adopts a multi-strategy and multi-source search approach taking into account Kitchenham's guidelines [26]. It is based on both automated and manual searches. We used search engines like Scopus, digital libraries such as IEEE Xplore Digital Library, ACM Digital Library, Springer Link, journal issues like *Empirical Software Engineering Journal* (EMSE), *Information and Software Technology* (IST), and conference proceedings from *Empirical Software Engineering and Measurement Conference* (ESEM), *Evaluation and Assessment in Software Engineering Conference* (EASE) and *International Conference on Software Engineering* (ICSE). The intention is to achieve exhaustive coverage.

Table 3 presents the sources used in the search strategy for each one of the secondary studies. Although the Systematic Review took place a year and a half after the Rapid Review, they both covered studies published on the same time span. This tries to guarantee a fair comparison. We are aware that more search sources may generate more evidence. However, a Rapid Review proposes the omission/abbreviation of traditional Systematic Review steps. And, our primary goal is identifying the difference between both and what we are losing or gaining from this omission/abbreviation.

The Rapid Review's and Systematic Review's search strings are also different. However, none of the convergences and divergences between them are intentional. The protocols were defined independently, as we already mentioned. When two different people conduct two secondary studies about the same topic, separately, it is expected that parts of the protocol are somewhat different, including the search strings. That kind of scenario is exactly what we want to reproduce.

If we used the same Rapid Review search string on the Systematic Review, it would neither be a fair nor a real scenario comparison. Rapid Review's search string is

Table 2
Secondary Studies Search Strings.

RR Search String	SR Search String
(customer OR "product owner" OR stakeholder) AND (collaborat* OR participat* OR cooperat* OR relation* OR involvement OR engagement) AND ("software development" OR "software engineering" OR "software project")	(customer OR client OR "product owner") AND (collaboration OR participation OR contribution OR cooperation) AND ("software engineering" OR "software development")

Table 3
Secondary Studies Search Sources

RR Search Sources	SR Search Sources
Automatic Search Scopus	Automatic Search ACM DigitalLibrary, IEEEExplorer Digital Library, Scopus, Springer Link
	Manual Search EASE, EMSE, ESEM, ICSE, IST

supposed to be defined in close collaboration with experts on a tight schedule. Systematic Review's search string, on the other side, is usually defined based on the author's expertise and with plenty of time for calibration and pondering. Those are two very different settings.

Even though, the Rapid Review and Systematic Review occurred to have the same main terms (<customer> AND <collaboration> AND <software engineering>), the synonyms for each term are slightly different. Table 2 presents the strings of the secondary studies used to search in the automated search sources.

The Rapid Review's search process returned 1,973 papers, while the Systematic Review returned 12,297 papers (11,383 non-duplicated), of which 3,601 came from the automatic search and 8,696 from the manual search, without excluding duplicates. The difference between the Rapid Review and Systematic Review is very significant, but most of the studies found in the Systematic Review came from the manual search. This happens because, to have high coverage, we considered all studies published in all journal issues and conference proceedings of the five Systematic Review's manual search sources.

4.7. Selection Procedure

Figure 2 depicts the selection procedure of the Rapid Review and the Systematic Review side-by-side.

The Rapid Review selection procedure³ was conducted by a single researcher in three steps. In the first step, the researcher analyzes only the title of the 1,973 papers returned by the search, leaving 84 papers after excluding those clearly out of the RR's scope. In the second step, the researcher excludes the papers that remained from the previous step based on their abstracts, leaving 47 papers. From those papers, five could not be found. The leading researcher did

³The RR selection procedure is available at <http://bit.ly/2su8xC8>

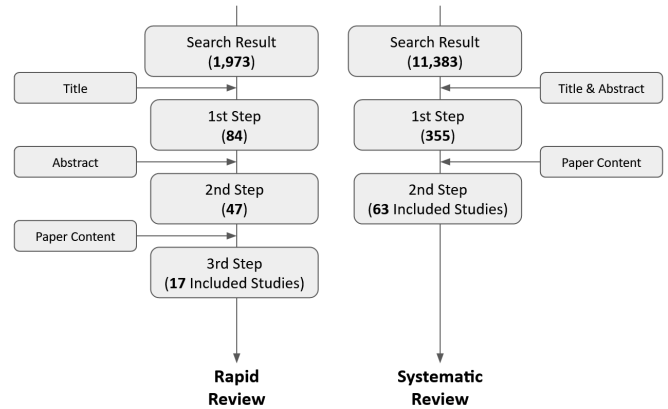


Figure 2: Selection procedure of both secondary studies.

not try to contact the papers' authors due to Rapid Review's time constraints. So, those five papers were excluded before the next step. Finally, in the third and last step, the researcher excludes the remaining papers by reading their entire content, leaving only papers that were included in the Rapid Review, which are 17. In addition, no quality appraisal is conducted; consequently, no paper is excluded due to it.

On the other hand, the Systematic Review selection procedure⁴ comprised two steps plus the quality appraisal. In the first step, we analyzed the title and abstract of the 11,383 non-duplicated papers, resulting in the selection of 355 papers.

In the first selection step of the Systematic Review both the papers' titles and abstracts were analyzed. In theory, that would reduce the probability of unduly excluding papers, in contrast to what may occur in the first step of Rapid Review's selection procedure, which selects based on the paper's title only. That assumption has proved to be true, at least in our case; we discuss the results in Section 6. This first selection step also passed through a validation involving four researchers grouped into three pairs. The first author paired with the others. Due to the high number of studies, we conducted the validation in pairs for 10% of the papers (1,139), which were randomly chosen. We verified the agreement of pairs with Kappa statistics [29], obtaining two pairs with a moderate agreement and one with a substantial agreement. The remaining papers were conducted by the first author only.

⁴The SR selection procedure is available at: <https://bit.ly/3XXjFVn>

Similar to the Rapid Review study, we also could not find the documents with the full content of some of the Systematic Review's papers, 29 to be more precise. However, unlike with the Rapid Review, in the Systematic Review, we tried contacting the papers' authors. We have succeeded in obtaining 12 out of the 29 papers.

In the second selection step of the Systematic Review, we conducted the paper analysis fully in pairs to reduce selection bias. The papers were excluded analyzing their entire content, resulting in the selection of 63 papers.

The next step (quality appraisal) does not exactly fit in the selection procedure. No paper was excluded, which makes the Systematic Review end up with 63 included studies.

Now we explore how the Rapid Review and the Systematic Review converge and diverge regarding their included papers. Figure 3 depicts a Venn Diagram to represent the quantitative of both the secondary studies' included papers.

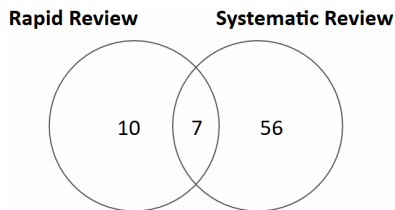


Figure 3: The number of papers included in the Rapid Review and the Systematic Review.

One would probably expect that the Rapid Review's included papers would be a subset of the Systematic Review's. However, it has been shown to be false in our case. Following, we explore the reasons why some papers (10) were included in the Rapid Review but not in the Systematic Review:

- **Absence of terms on the Systematic Review's search string [8 papers]:** Most of the papers that were included in the Rapid Review but not in the Systematic Review occurred due to the absence of the term "stakeholder" on the Systematic Review's search string. That term is present on the title and/or abstract of those papers, that were not even returned by the Systematic Review search procedure.
- **Selection bias during the 1st step of the Systematic Review's selection procedure [1 paper]:** Only one paper was excluded from the Systematic Review during the first selection procedure step (title and abstract) but included in the Rapid Review. The only reason we see to explain that is selection bias.
- **Selection bias during the 2nd step of the Systematic Review's selection procedure [1 paper]:** Just one paper excluded in the Systematic Review during the second selection procedure step (paper content) but included in the Rapid Review. The reason to explain that is selection bias, as in the item before.

Now we explore the other way around. The reasons why some papers (56) were included in the Systematic Review but not in the Rapid Review:

- **Absence of some search sources on the Rapid Review [14 papers]:** Those papers were not even returned by the Rapid Review search procedure based on Scopus only. In the Systematic Review they were found in other search engines such as the ACM Digital Library, IEEEExplorer, and SpringerLink.
- **Selection bias during the 1st step of the Rapid Review's selection procedure [38 papers]:** Most of the papers included in the Systematic Review but not in the Rapid Review were returned by the Rapid Review's search source. However, they were excluded during the first phase of the Rapid Review's selection procedure (title only) due to selection bias. Excluding papers based on title only may have influenced this number.
- **Selection bias during the 3rd step of the Rapid Review's selection procedure [4 papers]:** The reason to explain these unduly exclusion in this step (paper content) is selection bias.

4.8. Quality Appraisal

Although there is a section about quality appraisal in the Rapid Review protocol, it only states that it does not conduct such kind of procedure.

In contrast, we conduct a quality appraisal of the primary studies in the Systematic Review. We followed the 11 criteria method by Dyba and Dingsoyr [30]. The studies were given scores for each criterion. The quality appraisal was performed by a single researcher and reviewed by another one. In the end, no paper was excluded based on the quality assessment.

4.9. Extraction Procedure

The Rapid Review extraction procedure was performed by a single researcher. The extraction spreadsheet has the following information about the included papers: metadata about the papers; and evidence from the papers that answers the research questions.

The Systematic Review extraction procedure was performed by a single researcher and reviewed by another one. The extraction spreadsheet contains not only the same data as the Rapid Review — metadata about the papers and evidence from the papers that answer the research questions — but also reference and contextual data from the included papers.

4.10. Synthesis Procedure

Both the Rapid Review and the Systematic Review adopted thematic analysis as a synthesis method. This is just a coincidence since the procedural decisions of the secondary studies are independent. The nature of the collected data (i.e., qualitative) also reduces the space of possibilities. The Rapid Review synthesis was conducted by a single researcher only. In contrast, the Systematic Review

synthesis was conducted by a single researcher and reviewed by another one.

Despite both secondary studies adopting the same synthesis method, the guidelines they followed are different. The Rapid Review adopted Fereday and Cochrane's [31] guidelines, which define the following steps: familiarization with the data, creation of initial codes, search for themes, review of themes, and construction of the report. While the Systematic Review adopted the guidelines of Cruzes et al. [32] and Braun and Clarke [33], which comprises the following steps: reading data, identifying codes, forming themes, reviewing themes, and subsequently, building theme descriptions.

4.11. Report

The Rapid Review's findings are reported through an Evidence Briefing [34], which is a one-page document that presents the main findings of an empirical research. The goal of such kind of medium is to present the results to experts more appealingly and straightforwardly compared to the traditional research paper format. In addition, there was a workshop together with the experts involved in the Rapid Review to present the Evidence Briefing and discuss the findings. According to them, the experts mentioned that, although the Evidence Briefing was easy to read, they could not understand some points just reading it. However, those points became clear during the workshop discussion. Still, regarding the Rapid Review's Evidence Briefing, it has a URL to access the list of primary studies that composed the Rapid Review. Nonetheless, there is no way in the Evidence Briefing to know from which primary study a specific finding comes from, which is a downside.

4.12. Findings

In this section, we present how the findings of the Rapid Review and the Systematic Review converge and/or diverge, and why. To explore the differences, we have to mention some of the concepts related to the topic of the secondary studies, which is "customer collaboration" in software development projects. However, we do not dive into details — as the Rapid Review and Systematic Review do — since this is not the goal of this paper.

The Rapid Review identified 30 findings, summing up all the findings that answer each of the four research questions. The Systematic Review, coincidentally, has identified the same number of findings. However, the findings are not all the same. 12 out of the 30 Rapid Review's findings are not identified in the Systematic Review, while 16 out the 30 Systematic Review's findings are not identified in the Rapid Review.

The relation between the secondary studies' findings is N-N. So, a finding of one of the secondary studies may correspond to more than one finding of the other. To illustrate, take a look at Figure 7.

Another peculiarity is that, in some cases, a finding does not immediately seems to be covered by another. For example, take a look on Figure 7. The finding "Risk assessment", which answers Rapid Review's RQ4 about strategies

to improve customer collaboration, matches the Systematic Review's finding "Use of agile methodology". They do not look the same at first sight. But the latter comprises the former when reading the Systematic Review's findings in detail, which is out of the scope of this paper. This occurs because, in general, the findings of the Systematic Review are more high level compared to the Rapid Review's ones, sometimes covering more than one Rapid Review finding. In Section 6 we discuss this occurred.

Following there the reasons why some findings were identified on one secondary study but not on the other, and vice-versa:

- **Divergence due to the search strategy [Rapid Review 1 finding and Systematic Review 5 findings]:** In this case, the papers that originated the findings on one secondary study were not even returned by the other's search. This is a coverage problem on the search strategy. Either due to differences between search sources or search strings. This situation occurred with five findings that were identified in the Rapid Review, but not on the Systematic Review. While just one finding was identified in the Systematic Review, but not on the Rapid Review for that same reason.
- **Divergence due to the selection procedure [Rapid Review 11 findings and Systematic Review 1 finding]:** In this case, the papers that originated the findings on one secondary study were returned by the other's search. However, those papers were unduly excluded during the selection procedure. This situation occurred with just one finding that was identified in the Rapid Review, but not on the Systematic Review. On the other side, 11 findings were identified in the Systematic Review, but not on the Rapid Review for that same reason.
- **Divergence due to the synthesis procedure [Rapid Review 4 findings and Systematic Review 6 findings]:** In this case, the papers that originated the findings on one secondary study were returned by the other's search, and even included after the selection procedure. However, the findings were not identified in the synthesis procedure of the latter. This situation occurred with six findings that were identified in the Rapid Review, but not on the Systematic Review. While four findings were identified in the Systematic Review, but not on the Rapid Review for that same reason.

Following we present, in more detail, the differences between the findings of the Rapid Review and the Systematic Review per each research question.

Differences between the findings of the secondary studies regarding RQ1 - Benefits of customer collaboration

The Rapid Review identified nine benefits of customer collaboration, while the Systemic Review found six. Three benefits were identified in both the secondary studies. The others are disjoint. Figure 4 shows the benefits identified on

both the Rapid Review and the Systematic Review, side-by-side.

There are two reasons why some findings that answer RQ1 were identified in the Rapid Review but not in the Systematic Review: divergence due to the search strategy (4 findings), i.e. the search string could not find one or more papers that originated the findings; and divergence due to the synthesis procedure (2 findings), i.e. the papers that originated the findings were found and included, but the findings could not be identified during the synthesis procedure. While, on the opposite side, the reasons why some findings that answer RQ1 were identified in the Systematic Review but not in the Rapid Review are: divergence due to the selection procedure (3 findings), i.e. papers were unduly excluded in the first selection step.

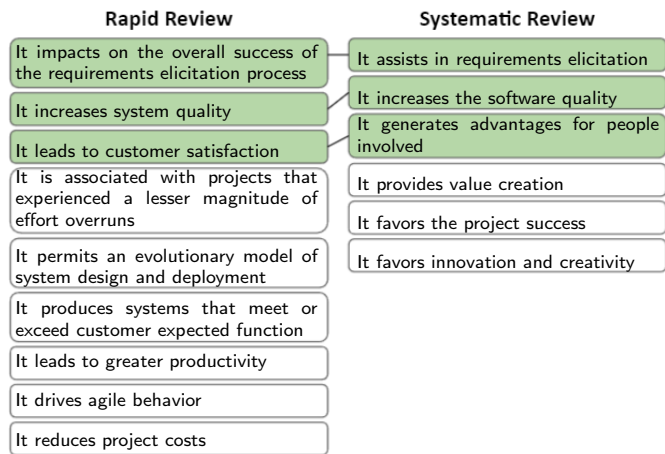


Figure 4: Comparison of customer collaboration benefits (RQ1) identified on the secondary studies.

Differences between the findings of the secondary studies regarding RQ2 (Problems caused by low customer collaboration)

The Rapid Review identified five problems caused by low customer collaboration, while the SR found six. Four problems identified on the Systematic Review are not covered by the Rapid Review, while there are only two the other way around. Figure 5 shows the problems identified on both the Rapid Review and the Systematic Review, side-by-side.

The reason some findings that answer RQ2 were identified in the Rapid Review but not in the Systematic Review is the divergence due to the synthesis procedure (2 findings). While the reason some findings that answer RQ2 were identified in the Systematic Review but not in the Rapid Review is the divergence due to the selection procedure (4 findings).

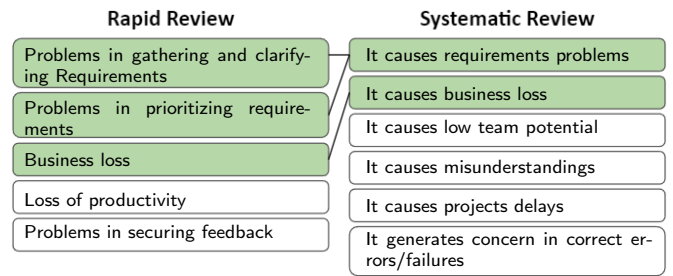


Figure 5: Comparison of problems caused by low customer collaboration (RQ2) identified on the secondary studies.

Differences between the findings of the secondary studies regarding RQ3 (Challenges to establish customer collaboration)

The Rapid Review identified five challenges to establish customer collaboration, while the Systematic Review found eight. Three challenges were identified in both the secondary studies. Figure 6 shows the challenges identified on both the Rapid Review and the Systematic Review, side-by-side.

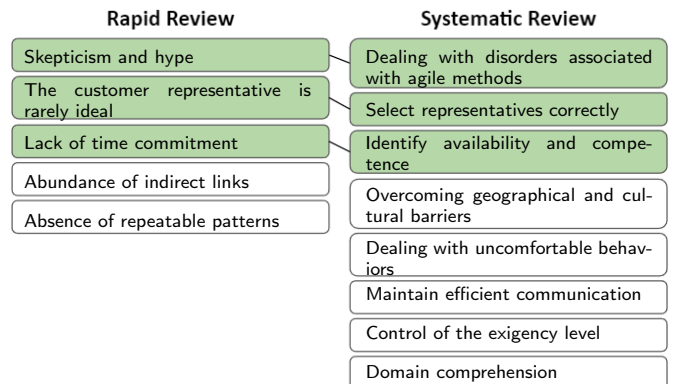


Figure 6: Comparison of challenges to establish customer collaboration (RQ3) identified on the secondary studies.

The reasons why some findings that answer RQ3 were identified in the Rapid Review but not in the Systematic Review are the divergence due to the selection procedure (1 finding) and the divergence due to the synthesis procedure (1 finding). While the reasons why some findings that answer RQ3 were identified in the Systematic Review but not in the Rapid Review are the divergence due to the selection procedure (2 findings) and the divergence due to the synthesis procedure (3 findings).

Differences between the findings of the secondary studies regarding RQ4 (Strategies to improve customer collaboration)

The Rapid Review identified 11 strategies to improve customer collaboration, while the Systematic Review found

10. Despite that, five strategies identified on the Systematic Review are not covered by the Rapid Review, while three are not covered the other way around. Figure 7 shows the strategies identified on both the Rapid Review and the Systematic Review, side-by-side.

The reasons some findings that answer RQ4 were identified in the Rapid Review but not in the Systematic Review are: divergence due to the search strategy (1 finding) and divergence due to the synthesis procedure (1 finding). Meanwhile, the reasons why some findings that answer RQ4 were identified in the Systematic Review but not in the Rapid Review are: divergence due to the search strategy (1 finding), divergence due to the selection procedure (2 findings), and divergence due to the synthesis procedure (1 finding).

Finally, Figure 8 shows a general picture of the main quantitative data comparing the Rapid Review with the Systematic Review.

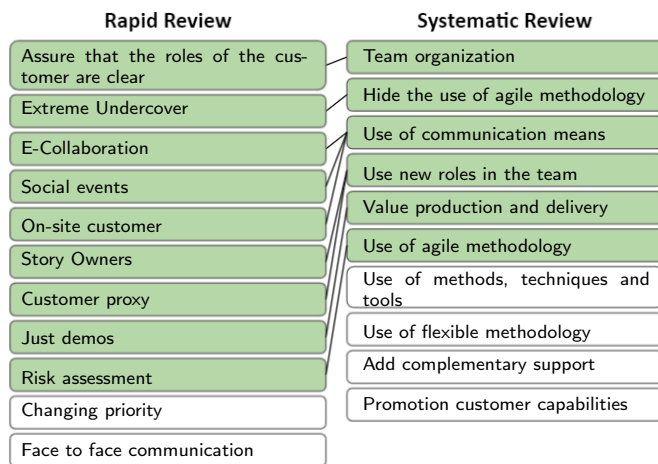


Figure 7: Comparison of strategies to improve customer collaboration (RQ4) identified on the secondary studies.

5. Results of the Experts Survey

We had 39 participants in our survey. Regarding the age of the participants, we had a very diverse sample, ranging from 20 to 55 years of age. This contributes to a broad and diverse perspective. In Figure 10, it is possible to observe the full distribution of ages. The average age is 36 years, and the median is also 36 years.

Concerning the participants' educational attainment, ~36% of them have a Ph.D. degree, followed by an undergraduate degree (~28%), a master degree (~23%), and a high school degree (~13%). Also, most of the participants are from Industry (~54%), followed by those who work in both Academy and Industry (~26%) and Academy Only (~20%).

The participants' current role are very diverse, so we grouped them. Leadership/Management are ~44% of our sample, followed by Professors/Researchers with ~28%, and Engineers/Developers, with ~26%. About 2% preferred not to share.

Regarding the participants' professional experience in years, we also had a very diverse sample. The sample ranges from 1 to 32 years of experience, with an average of 13 years of experience and a median of 14 years of experience. Figure 11 shows the trends in participants' professional experience in years.

One may believe that older and/or more experienced experts would be more critical and, in consequence, would assign lower relevance to the secondary studies findings. Thus, we ran Spearman's rank correlation tests. One, between the participants' age and the level of relevance they assign to the findings of both the secondary studies ($\rho = 0.0804$); and another considering the participants' professional experience and also the level of relevance they assign to the findings of both the secondary studies ($\rho = 0.0684$). We could not find significant correlation in any case.

That being said, our main hypothesis is that the participants would attribute a lower level of relevance to the Rapid Review's findings when compared to the Systematic Review's findings, even though they do not know from where (Rapid Review or Systematic Review) the pieces of evidence come from.

Therefore, we performed a two-tailed Wilcoxon signed-rank hypothesis test. Again, we could not find significant difference ($W = 567795$, $p\text{-value} = 0.07129$), between the relevance attributed to the findings of Rapid Reviews and Systematic Reviews. Despite the fact that the $p\text{-value}$ does not enable us to reject the null hypothesis, its value is quite close to the threshold (0.05). So, we generated a violin plot with the data, in order to analyze not only with inferential but also with descriptive statistics. Figure 12 presents the violin plot that reveals data shape and dispersion.

As one can see, the relevance of the findings from Rapid Review and Systematic Review appears to be quite similar, both more concentrated on the positive side. However, a slight difference can be observed when plotting the data as a stacked bar chart, as shown in Figure 13. The Systematic Review has a few more positive values, and less neutral and negative values, although the difference is clearly negligible.

When we look at each piece of evidence separately, as we can see in Figure 14, one can note that the top 10 most relevant findings (slightly agree + strongly agree), according to the respondents, are a mix of findings shared by both studies (Rapid Review and Systematic Review) and the evidence found by Systematic Review only. Five pieces of evidence come from both studies, and five from the Systematic Review only.

Additionally, the top ten least relevant findings, according to the respondents, are composed of two pieces of evidence from both studies, four findings from the Rapid Review, and four from the Systematic Review. The evidence with the worst evaluation among all is shared by both the Rapid Review and the Systematic Review.

Comparing a Rapid Review with a Systematic Review

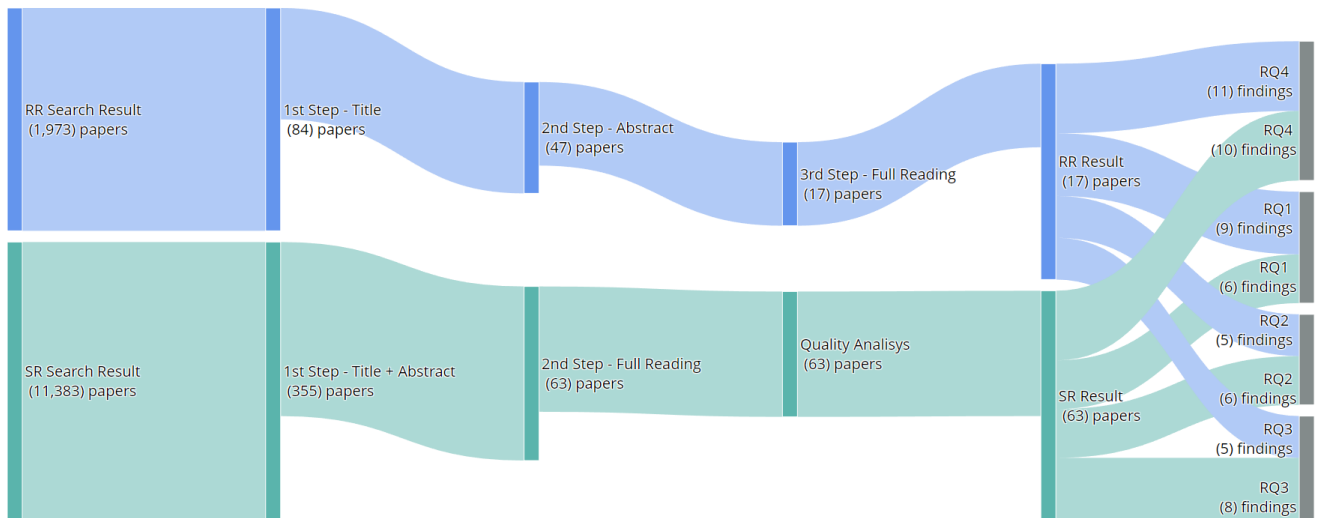


Figure 8: Procedures of both the secondary studies.

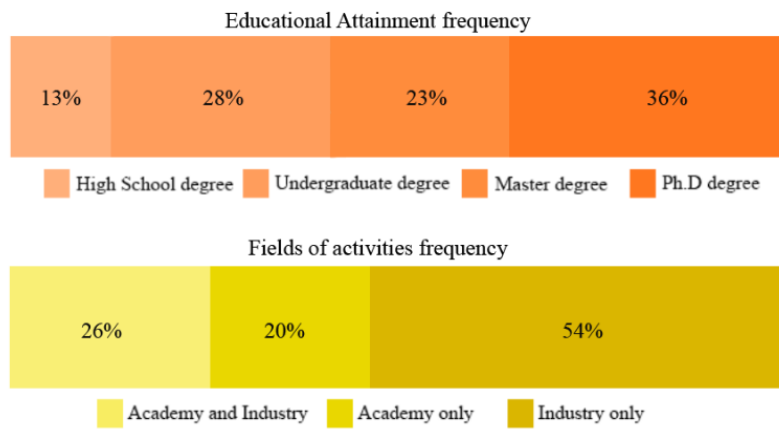


Figure 9: Participants survey demographic data.

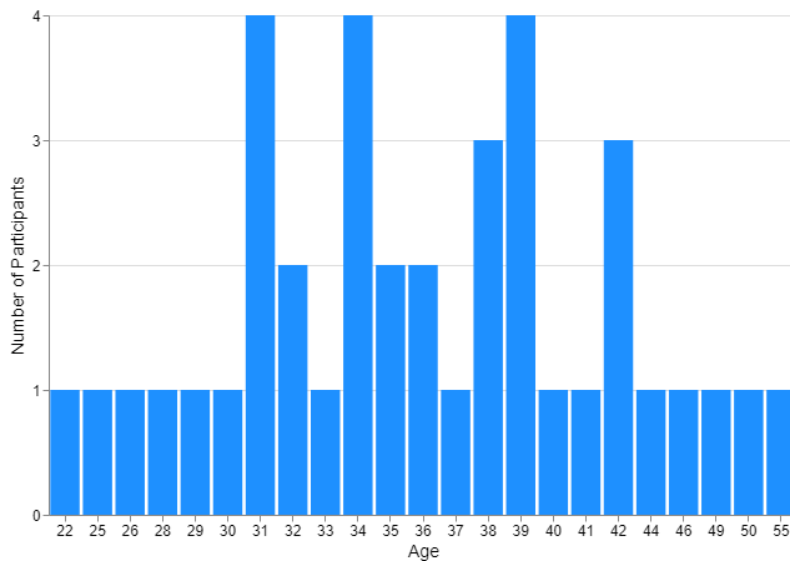


Figure 10: Age Distribution of Participants.

Comparing a Rapid Review with a Systematic Review

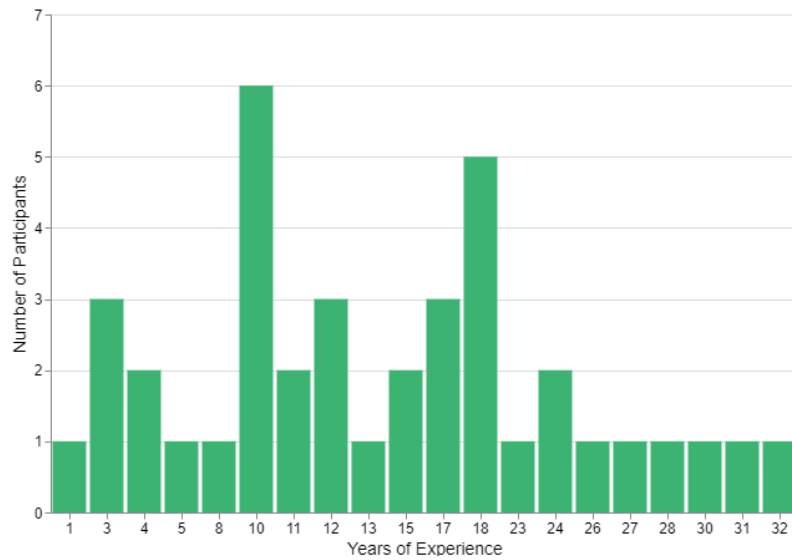


Figure 11: Experience of Participants.

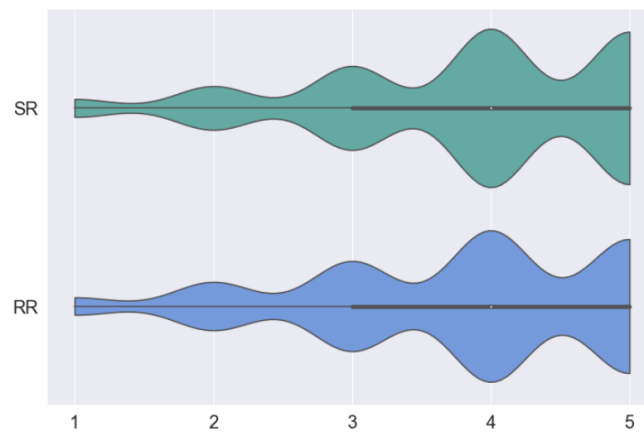


Figure 12: Shape and dispersion of the level of relevance the experts assigned to the SR's findings compared to the RR's findings.

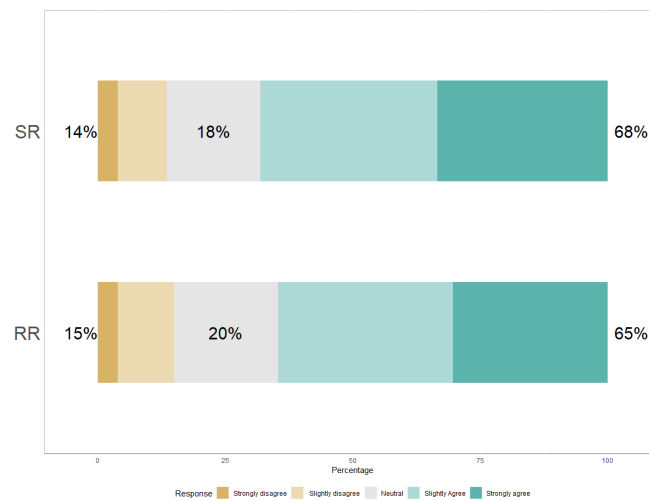


Figure 13: Level of relevance the experts assigned to the SR's findings compared to the RR's findings.



Figure 14: Level of relevance the experts assigned for each of the secondary studies' findings.

6. Discussion

IRQ1 What factors explain the difference in effort between a Systematic Reviews and a Rapid Reviews?

To understand which factors explain the difference between the efforts undertaken in each method, we need to revisit some of their processes.

Regarding the protocol, the Systematic Review's one is more detailed, which fosters replication and scrutiny. On the other hand, the Rapid Review protocol avoids theoretical justifications that might not be relevant to experts, saving their scarce time.

Regarding the search strategy, the Rapid Review seems to have a better cost-benefit when compared to the Systematic Review, at least under a quantitative perspective. Most (~78%) of the papers included in the Systematic Review (63) were returned by the Rapid Review search (49). Even though the Rapid Review search returned one order of magnitude fewer papers (1,973 vs. 11,383) than the Systematic Review. This, in turn, cuts off much of the effort spent on Rapid Review's selection procedure, while still guaranteeing a considerable similarity of relevant papers with the Systematic Review. The manual search was a great villain in this context. Just two out of 8,696 papers from the manual search ended up being included in the Systematic Review. The manual search is also responsible for more than 90% of the four months spent executing the entire Systematic Review search strategy. In the end, a manual search strategy seems to be not promising, even for a Systematic Review, unless the topic is critical and exhaustive coverage is mandatory.

Regarding the selection procedure, one important aspect to take into consideration is that the Systematic Review selection procedure took six months, while the entire Rapid Review was conducted in six days. Here we are in a trade-off situation that choosing an option is a matter of pondering the research goals and the target audience of the secondary study. The number of papers excluded during the Rapid Review selection procedure may be high for a rigorous academic-focused Systematic Review. However, on the other side, six months and nine researchers to conduct just the Systematic Review selection procedure may also be unacceptable from expert' perspective.

We can consider that the difference between the effort made in both secondary studies was mainly in the search strategy and the selection procedure, which directly reflects on the time frame. Regarding the time frame, the difference is huge. The Systematic Review took one year and two months, while the Rapid Review took only six days. The main bottleneck on the Systematic Review is concentrated in two activities: executing the search strategy, and the selection procedure. Together they took ten months. Those activities are also less intellectual when compared to others. So, researchers spend a lot of time and effort in a Systematic Review with activities that do not explore their full potential. Experts also cannot wait for months and sometimes years to make their decisions in real-world scenarios.

Therefore, the time and cost to conduct a Systematic Review can be prohibitive in experts' contexts. Thus, at least in such situations, a Rapid Review may be an adequate choice. Moreover, a Rapid Review may be executed in the experts' context as a previous low-cost step before deciding to invest in a high-cost Systematic Review.

IRQ2 What factors explain the difference in relevance between the findings of a Systematic Review and a Rapid Review?

We first expected that we would find large differences between the findings of the Rapid Review and the Systematic Review. This assumption was based on the common belief that exhaustive coverage would consequently be more robust. However, when carrying out our comparison, this assumption was questioned. So we decided to carry out a survey as a way to deeply investigate how relevant the findings of the Rapid Review and Systematic Review are from the perspective of experts in industry and academia.

Again, we were surprised to find that the relevance of the findings from the Rapid Review and Systematic Review appears to be quite similar, both more concentrated on the positive side in the Systematic Review.

With the survey results, we observed that there is no significant difference regarding the relevance of the findings from the Systematic Review when compared to those from the Rapid Review. This may contradict the initial belief that evidence from Systematic Reviews is necessarily more relevant than evidence from Rapid Reviews.

One factor that we can consider as potentially decisive in the in relevance between the findings of each method seems to have been the synthesis procedure. When analyzing the evidence that stood out positively in the Systematic Review, we observe that the findings are more coarse-grained, while those of the Rapid Review are more fine-grained. This may have influenced the assignment of relevance by the experts.

Thus, considering such an expressive difference between Rapid Review's and Systematic Review's time frames, and the fact that the differences between the findings are quite balanced, we believe the software engineering research community should accept Rapid Review as a valid secondary study method, among the others we already have. At least when the goal is delivering evidence promptly to experts.

IRQ3 Are the findings from a Systematic Review and a Rapid Review significantly different in some ways?

The Rapid Review and the Systematic Review are inherently different and complementary; consequently, their findings were different in some ways.

Due to the presence of one particular term (i.e., stakeholder), the Rapid Review could include ten papers that were not even found by the Systematic Review search string. This had a significant impact on the Systematic Review findings. Thus, a search strategy cannot be reduced to the search

sources only. The search strings also can play an important role.

The selection process has shown to be the Achilles' heel of the Rapid Review. Although the Rapid Review search could find 49 of the 63 papers included in the Systematic Review, 42 of them were unduly excluded during the Rapid Review selection procedure. Most of those unduly exclusions (38 out of 42) occurred during the first step of the Rapid Review selection procedure, where the researcher excluded papers based on papers' titles only. We took a step further and investigated how the unduly paper exclusions impacted the findings of both secondary studies. And again, it shows us that the Rapid Review lost a fair number of findings compared to the Systematic Review, due to divergences in the selection procedure.

Regarding the synthesis procedure, this is probably the most subjective activity of a secondary study, even when one follows a systematic method like thematic analysis. Despite such subjectivity, the number of findings identified by one secondary study but not by the other, due to the synthesis procedure, is quite similar. The Rapid Review could not identify 4 out of the 30 findings of the Systematic Review due to divergences in the synthesis procedure. While the Systematic Review could not identify 5 out of the 30 findings of the Rapid Review, for the same reason.

But when we consider the content of the findings, while the Systematic Review identified more coarse-grained findings – sometimes even comprehending more than one Rapid Review finding – the Rapid Review identified more fine-grained ones, which makes the Rapid Review's findings more specific and straightforward. This may have happened because the expert' influence guided the Rapid Review towards more specific findings, but it is just a conjecture for future work. Furthermore, it is also necessary take into consideration that the Systematic Review' findings are presented more in-depth than the Rapid Review' findings, with descriptions and contexts, and this can also be beneficial to experts.

6.1. Additional discussions

Regarding the obvious question that comes to anyone's mind — should we conduct Rapid Reviews or not? — we believe it is also obvious that the answer is not definitive. It may vary according to different contexts. In particular, it lays on another question — is the difference between the Rapid Review and the Systematic Review acceptable? — which is also hard to answer and very context-dependent. All that being said, we believe a difference in time frame from six days (Rapid Review) to one year and two months (Systematic Review) is so high, and perhaps impractical in many industry scenarios, that Rapid Reviews should at least be considered in such contexts. Additionally, the positive perceptions of experts towards Rapid Reviews reported by Cartaxo et al. [7] also reveal that experts are willing to cope with any threat to validity inherent to Rapid Reviews, as long as they are aware of them. The findings of both secondary studies demonstrate that Systematic Reviews and

Rapid Reviews can bring relevant contributions to professional practice. Also, they can be complementary since we could find unique pieces of evidence with both of them. The context will define which method can be applied.

If we suggest a more specific context for its use, we can say that a Rapid Review can be adequate when there is a need to solve an urgent problem but established on a solution-oriented and based on scientific evidence. A Systematic Review can be produced when there is time for it, and there is a need to impact the production in the long term of any problem identified in the industry or in case of critical systems.

Finally, one should consider all the results of this research with due care. First, we compare just one Rapid Review with one Systematic Review. Many other comparisons need to arise until we are confident about the real impact of flexibilizing Systematic Review's method. Second, both Rapid and Systematic have high variability. So, a Rapid Review or a Systematic Review that adopts different strategies regarding any of their procedures might reveal a different impact on their results. For instance, a Rapid Review that does not apply any systematic synthesis method would probably exhibit considerably different findings compared to the Rapid Review we analyzed in this research, which adopted thematic analysis. And third, as much systematic a secondary study may be, it still inherently exhibits some level of subjectivity in most of its activities. Many aspects may interfere with the results.

7. Conclusions

This paper shows that the time frame to plan, perform, and report a Rapid Review compared to a traditional Systematic Review is huge. Six days for the former and one year and two months for the latter. We also could observe the main bottlenecks on Systematic Review are executing the search strategy and the selection procedure. Rapid Review's search strategy also has been shown to be more cost-effective than the Systematic Review. It returned most of the papers included in the Systematic Review but taking considerably less time and effort. On the other side, the selection procedure has shown to be the Rapid Review main drawback. Many papers that could be included were unduly excluded during the Rapid Review's selection procedure. In the end, both the Rapid Review and the Systematic Review could identify the same number of findings to answer their shared research questions, but not without some differences.

We emphasize with these results that Rapid Review and Systematic Review are inherently different and complementary. However, more comparisons are needed to draw stronger conclusions about the differences between the two methods. This is the first step towards other initiatives alike. As soon as there are more Rapid Reviews out there, we plan to compare them with pre-existent Systematic Reviews. We also believe it is viable to have a comparison between two pre-existent Rapid Review and Systematic Review, even when they start from different research questions. As long as

they share some characteristics, it is possible to have partial comparisons that together may accumulate a relevant body of knowledge around the actual differences between Rapid Reviews and Systematic Reviews, as there are in medicine.

7.1. Limitations and Threats to Validity

The Rapid Review was conducted by the same research group that conducted the Systematic Review. This inherently exhibits desired and not desired issues. The desired one is that researchers in our research group deeply participated in the Rapid Review, so they could provide all the details that otherwise would be hard to discover just by reading the original publication. On the other side, the fact that our research group conducted the Rapid Review may introduce bias when conducting the Systematic Review. To mitigate this bias, as mentioned in Section 3.2, the leading researcher of the Systematic Review did not read or had any contact with the original Rapid Review in any way, at least not until the end of the entire Systematic Review.

Regarding the comparison between the Rapid Review and the Systematic Review, there is at least one situation that may bias the results we obtained. The Rapid Review leader had slightly more experience conducting secondary studies when compared to the leader of the Systematic Review. So, we cannot affirm if or how this fact impacts the time frame and other steps of the secondary studies.

In addition, a limitation that needs to be considered is that we only compare one Rapid Review with one Systematic Review. Thus, there is a need for more comparisons to draw stronger conclusions about the differences between Rapid Reviews and Systematic Reviews, but this is the first step towards other initiatives alike.

Regarding the survey, we followed the guidelines proposed to increase the validity of surveys in Software Engineering [35]. First, we defined what we wanted to evaluate – the pieces of evidence provided by the secondary studies – and then we created a draft survey. Next, we conducted a pilot with three researchers and based on their feedback, we fixed and improved the survey.

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CRedit authorship contribution statement

Caroline Pena: Leading, conceptualization, execution, writing, and revising. **Bruno Cartaxo:** Supervising, conceptualization (comparison and survey), execution (comparison

and survey), writing, and revising. **Igor Steinmacher:** Conceptualization (survey), and revising. **Deepika Badampudi:** Execution (SR), and revising. **Deyvson da Silva:** Execution (SR). **Williby Ferreira:** Execution (SR). **Adauto Almeida:** Execution (SR). **Fernando Kamei:** Execution (SR). **Sérgio Soares:** Supervising, conceptualization (comparison and survey), and revising.

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